QUANTUM PHYSICS

Q1. Why is the wave nature of matter not visible in our daily life.
(M.U. Dec 2009) [3 Marks]
Any matter particle of mass 'm' and velocity 'v' is associated with a wave of wavelength

$$\lambda = \frac{h}{mv} = \frac{h}{p}$$

where p = mv is the momentum of the particle.

The matter waves are not associated with
(i) a rest particle for which $\upsilon = 0$ and $\lambda = \infty$
(ii) a macroscopic particle for which m >> 0 and $\lambda \cong 0$.
This is the reason why wave nature of matter is not visible in our day to day life,

Q2. If an electron is accelerated at potential V, find the wavelength of matter wave associated with it. Give its importance.
(M.U. Dec 2008) [3 Marks]
Since the electron energy is

$$E = \frac{1}{2} m v^2 = eV$$

the wavelength associated with an electron wave can be written as

$$\lambda_T = \frac{h}{\sqrt{2mE}} = \frac{h}{\sqrt{2meV}} = \frac{12.25}{V}$$

The importance of this wavelength is that when V= 54 volts for which theoretical electron wavelength is found to be

$$\lambda_T = 1.67 \quad A^{\circ}$$

which is very close to the experimental value, λ_E.
This agreement confirms de Broglie's hypothesis of matter waves.

Q3. Explain de Broglie's Hypothesis of matter waves and deduce the expression for λ.
(M.U. May 2008; Dec 2010, 2011; June 2019) [3 Marks]
De Broglie introduced the concept of matter waves. He suggested that the wave particle duality of radiation should be extended to micro-particles as well.

From the theory of radiations, it is known that the energy of a photon, a zero mass particle is given by

$E = h\upsilon = h\frac{c}{\lambda}$ —①

where,
c = velocity of light
υ = frequency
λ = wavelength

On the other hand, according to Einstein's mass energy relation, the energy of a non-zero mass particle with mass 'm' is given by

$E = mc^2$ —②

from ① and ②,

$h\frac{c}{\lambda} = mc^2$

i.e. $h = p\lambda$ —③

Therefore,

$$E = \frac{1}{2}mv^2 = \frac{(mv)^2}{2m} = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}$$

Thus the value of λ is given by

$$\lambda = \frac{h}{\sqrt{2mE}}$$

Q4. What are properties of matter waves?
(M.U. May 2015; Dec 2018) [5 Marks]
1. These waves are neither mechanical nor electromagnetic waves, these are hypothetical waves.
2. Matter waves with different de Broglie wavelengths travel with different velocities whereas electromagnetic waves of all wavelengths travel with the same velocity c.
3. The de Broglie wavelength depends on the kinetic energy of the matter particle.

$$\lambda = \frac{h}{\sqrt{2mE}}$$

4. Matter waves travel faster than light.

The energy of a matter wave being combined with Einsteins mass energy formula, it can be written as,

$$h\upsilon = mc^2$$
$$\upsilon = \frac{mc^2}{h}$$

Hence, the wave velocity of the matter wave becomes

$$u = \upsilon\lambda = \frac{mc^2}{h}\lambda$$

It can also be written as,

$$u = \frac{mc^2}{h} \cdot \frac{h}{p} = \frac{mc^2}{mv} = \frac{c^2}{v} = c\left(\frac{c}{v}\right) > 1$$

Since no matter particle can travel at a velocity greater than c, v << c and thus (c/v) >> 1.

Q5. Explain phase velocity and group velocity of matter waves.

(M.U. May 2011; Dec 2016, 2018) [5 Marks]

Q6. What is the significance of wave function Ψ of matter waves.

(M.U. May 2011) [3 Marks]

Waves represent the propagation of a disturbance in a medium.

Since micro-particles exhibit wave properties, it is assumed that a quantity Ψ represents a de Broglie wave.

The quantity Ψ is called a wave function which describes the behavior of a matter wave as a function of position and time.

It has no direct physical significance as it is not an observable quantity. However, the wave function provides information about the probability amplitude of position, momentum and other physical properties of a matter particle.

Q7. Derive the expression for energy eigen values for free particles in one dimensional potential well.

(M.U. May 2015, 2018; Dec 2015, 2017) [5 Marks]

Applying boundary conditions we get

(i)Ψ(x) = 0 at x = 0 which reduces equation $\Psi(x) = Ae^{jkx} + Be^{-jkl}$ to

A+ B = 0, and —①

(ii) Ψ(x) = 0 at x = L with which the equation $\Psi(x)=Ae^{jkx}+Be^{-jkl}$ becomes

$Ae^{jkx}+Be^{-jkL}=0$ —②

Combining ① and ② of these we get,

$$A\left(e^{jkx}-e^{-jkL}\right)=0$$

$$2\ Aj\sin kL = 0$$

Since A ≠ 0, sin kL = 0

and kL = nπ, with n = 0, 1, 2, ...

Using $k=\frac{2\pi}{\lambda}$ in the above equation, it is obtained that

$L=\frac{n\lambda}{2}$ —③

The energy of the particle inside the potential well, where V = 0, is only its kinetic energy. Hence

$$E=\frac{1}{2}mv^2=\frac{(mv)^2}{2m}=\frac{p^2}{2m}=\frac{h^2}{2m\lambda^2}$$

which with the use of ③ becomes,

$$E_n=\ n^2\frac{h^2}{8mL^2}$$

with n = 1, 2, 3, ...

This also proves that

In quantum theory, the lowest available energy for the particle is the energy quantum and the higher energies are multiples of it.

Q8. Write Schrödinger's time dependent and time independent wave equations and state the physical significance of these equations.

(M.U. Dec 2016) [7 Marks]

The one dimensional time dependent Schrödinger's wave equation is given as

$$-\frac{\hbar}{2m}\cdot\frac{\partial^2\Psi(x,\ t)}{\partial x^2}+V\Psi(x,\ t)=j\hbar\frac{\partial\Psi(x,\ t)}{\partial t}$$

The one dimensional time independent Schrödinger's wave equation is given as

$$-\frac{\hbar^2}{2m}\cdot\frac{d^2(x)}{dx^2}+V(x)\Psi(x)=E\cdot\Psi(x)$$

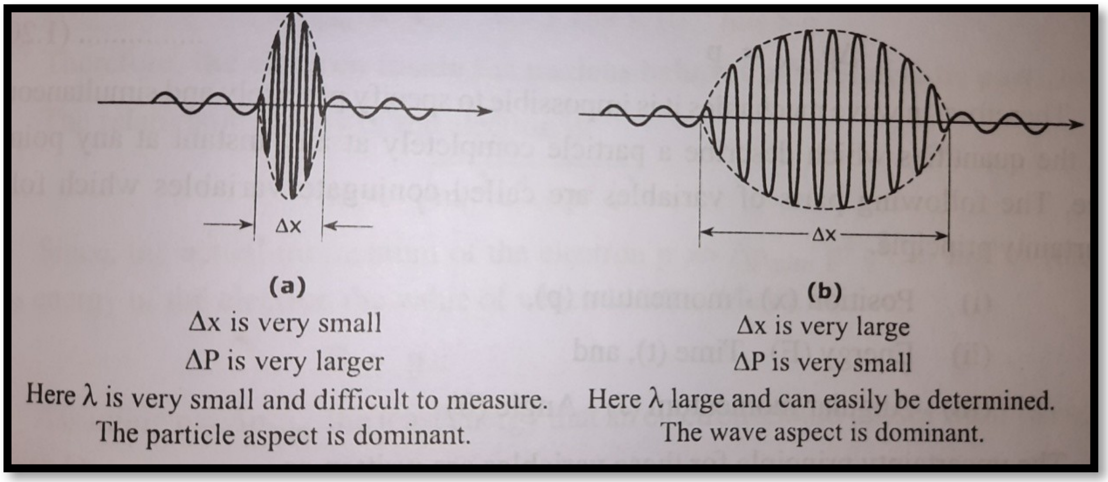
Q9. Explain Heisenberg's uncertainty principle with an example and give its physical significance.

(M.U. Dec 2008, 2009, 2013; May 2013) [5 Marks]

In wave mechanics a matter particle in motion is associated with a group of waves in the form of a wave packet.

When the wave packet is small, it is very easy to locate the particle but it is difficult to determine the wavelength (λ) associated with it and hence its momentum (p = h/λ) as shown in (a). In this case, particle characteristic of the matter wave is dominant over the wave characteristic.

On the other hand, if the wave packet is large the wavelength and hence the momentum of the particle can be easily determined but it is difficult to locate the particle which can be anywhere within the wave packet. This is shown in (b). In this case, the wave characteristic of the matter wave is dominant over the particle characteristic.



Therefore, it can be concluded that the position and momentum of a particle cannot be determined precisely and simultaneously. This is stated in the Heisenberg's Uncertainty Principle as ' the inaccurate Δx and Δp in the simultaneous determination of the position 'x' and momentum, 'p' respectively of a particle are related as

$\Delta x \cdot \Delta p \geq \hbar$

where, $\hbar = \frac{h}{2\pi}$, h being the Plank's constant.

It's physical significance is to determine and prove minimum energy possessed by an electron in the atom and the non existence electrons in a nucleus respectively.

Q10. Using Heisenberg's principle show that electrons cannot exist within the nucleus. (**M.U. May 2009, 2011, 2014, 2015, 2016; Dec 2012, 2014, 2017, 2018) [5 Marks]**

Initially assume that an electron is a part of nucleus.

The size of a nucleus is about 1 fermi = 10^{-15} m.

If an electron is confined within a nucleus, the uncertainty in its position must not be greater than the dimension of the nucleus, i.e., 10^{-15}m. Hence,

$$\Delta x_{max} = 10^{-15} m$$

From the limiting condition of Heisenberg's uncertainty principle, it can be written as,

$$\Delta x_{min} \cdot \Delta p_{min} = h$$

$$\Delta p_{min} = \frac{\hbar}{\Delta x_{min}} = \frac{6.63 \times 10^{-34}}{2 \times 3.14 \times 10^{-15}}$$

$$= 1.055 \times 10^{-19} \text{ kg-m/sec.}$$

Now, $$\Delta p_{min} = m \Delta v_{min}$$

Hence, $$\Delta v_{min} = \frac{\Delta p_{min}}{m} = \frac{1.055 \times 10^{-19}}{9.1 \times 10^{-31}}$$

$$= 1.159 \times 10^{11} \text{ m/s} > c$$

As $$\Delta v_{min} < v, \qquad v > 1.159 \times \frac{10^{11} m}{s} > c$$

Therefore, the electron inside the nucleus behaves as a relativistic particle.

The relativistic energy of the electron is

$$E = \sqrt{m_o^2 c^4 + p^2 c^2}$$

Since, the actual momentum of the electron p >> Δp_{min}, $p^2 c^2 \gg m_o^2 c^2$ (the rest mass energy of the electron the value of which is 0.511 MeV). Hence,

$$E = pc$$

Assuming p = Δp_{min} , the least energy that an electron should possess within a nucleus is given by

$$E_{min} = \Delta p_{min} \cdot c$$

$$= 1.055 \times 10^{-19} \times 3 \times 10^{8}$$

$$= 3.165 \times 10^{11} \text{ J}$$

Therefore,

$$E_{min} = \frac{3.165 \times 10^{-11}}{1.6 \times 10^{-19}} = 197 \quad MeV$$

In reality, the only source of generation of electron within a nucleus is the process of β - decay. The maximum kinetic energy possessed by the electrons during β - decay is about 100 KeV. This shows that an electron cannot exist within a nucleus.

Q11. Derive one dimensional time dependent Schrödinger for matter waves.
(M.U. May 2009, 2015, 2015, 2017; Dec 2012, 2013, 2014, 2015, 2017, 2018) [5/8 Marks]

For one dimensional case, the classical wave is described by the wave equation

$$\frac{d^2 y}{d x^2} = \frac{1}{v^2} \cdot \frac{d^2 y}{d t^2}$$

where y is the displacement, and v is the velocity of the wave travelling in a direction.

The displacement of the particle at any instant 't', at any point 'x' in space is given by the solution of equation of the normalization condition.

$$y(x,\ t) = A e^{j(kx - \omega t)}$$

where ω = 2πv and k = 2π / λ.
Here, ω is the angular frequency and k is the wave number.
In analogy with this, the wave function which describes the behaviour of the matter particle at any instant 't', at any point 'x' in space can be written as

$$\Psi(x,\ t) = A e^{j(kx\ -\ \omega t)}$$

where $$\omega = 2\pi v = 2\pi \frac{E}{h} = \frac{E}{\hbar}$$

and $$k = \frac{2\pi}{\lambda} = \frac{2\pi}{h} \cdot p = \frac{p}{\hbar}$$

The total energy of the particle is given by

E = Kinetic energy + Potential energy

$$= \frac{1}{2} m v^2 + V = \frac{(mv)^2}{2m} + V$$

Therefore, $$E = \frac{p^2}{2m} + V$$

Operating this on the wave function Ψ(x, t), it is found that

$$E\Psi(x,\ t) = \frac{p^2}{2m} \Psi(x,\ t) + V\Psi(x,\ t)$$

Differentiating equation $\Psi(x,\ t) = A e^{j(kx\ -\ \omega t)}$ with respect to 'x' and 't' it is obtained that

$$\frac{\partial^2 \Psi(x,\ t)}{\partial x^2} = -k^2 A e^{j(kx\ -\ \omega t)} = -k^2 \Psi(x,\ t)$$

$$\frac{\partial^2 \Psi(x,\ t)}{\partial x^2} = -\frac{p^2}{\hbar^2}\Psi(x,\ t)$$

Hence,
$$\frac{p^2}{2m}\Psi(x,\ t) = -\frac{\hbar^2}{2m}\cdot\frac{\partial^2 \Psi(x,\ t)}{\partial x^2}$$

Also,
$$\frac{\partial \Psi(x,\ t)}{\partial t} = -jA\omega e^{j(kx-\omega t)} = -j\omega\Psi(x,\ t)$$

Substituting ω,

$$\frac{\partial \Psi(x,\ t)}{\partial t} = -j\frac{E}{\hbar}\Psi(x,\ t)$$

$$E\Psi(x,\ t) = j\hbar\frac{\partial \Psi(x,\ t)}{\partial t}$$

Substituting the recent values of $\frac{p^2}{2m}\Psi(x,\ t) = -\frac{\hbar^2}{2m}\cdot\frac{\partial^2 \Psi(x,\ t)}{\partial x^2}$ and $E\Psi(x,\ t) = j\hbar\frac{\partial \Psi(x,\ t)}{\partial t}$ in $E\Psi(x,\ t) = \frac{p^2}{2m}\Psi(x,\ t) + V\Psi(x,\ t)$, it is obtained that

$$j\hbar\frac{\partial \Psi(x,\ t)}{\partial t} = -\frac{\hbar}{2m}\cdot\frac{\partial^2 \Psi(x,\ t)}{\partial x^2} + V\Psi(x,\ t)$$

or

$$-\frac{\hbar}{2m}\cdot\frac{\partial^2 \Psi(x,\ t)}{\partial x^2} + V\Psi(x,\ t) = j\hbar\frac{\partial \Psi(x,\ t)}{\partial t}$$

In the above equation, the first and second term on the left hand side represent the kinetic and potential energies respectively of the particle and the right hand side represents the total energy.
This is called the one dimensional time dependent Schrödinger equation.

Q12. Derive Schrödinger's Time Independent Schrödinger Equation.
(M.U. May 2016) [5 Marks]
If the potential energy of the particle varies only with the position of the particle and is dependent of time the field in which the particle is moving is called stationary for which

$$V = V(x)$$

In stationary fields, the time dependent part of the wave function is eliminated by using the method of separation of variables.
The wave function in this case is written as

$$\Psi(x,\ t) = \Psi(x)\phi(t)$$

Using this equation Schrödinger's time dependent equation can be written as

$$-\frac{\hbar^2}{2m}\cdot\frac{d^2\Psi(x)}{dx^2} + V(x)\Psi(x)\phi(t) = j\hbar\cdot\Psi(x)\cdot\frac{d\phi(t)}{dt}$$

Dividing this equation by the wave function equation , it is obtained that

$$-\frac{\hbar^2}{2m}\cdot\frac{1}{\Psi(x)}\frac{d^2\Psi(x)}{dx^2} + V(x) = j\hbar\frac{1}{\phi(t)}\cdot\frac{d\phi(t)}{dt}$$

The left hand side of the above equation is a function of x only and the right hand side is a function of only t and both the sides should be equal to a constant. With all possibilities the constant should be total energy. Hence

$$-\frac{\hbar^2}{2m}\cdot\frac{d^2\Psi(x)}{dx^2} + V(x)\Psi(x) = E\cdot\Psi(x)$$

This is called the one dimensional time independent wave equation.